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A compact and cost-effective gait simulator to advance prosthesis development with reduced reliance on human subject testing: Development, validation and application

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ABSTRACT

Gait simulators play a crucial role in assessing the performance of physical prototypes of prosthetic knees, validating numerical simulation findings, and reducing dependency on user trials during prosthesis development. However, their practical application is limited because of substantial capital investment required for sophisticated high degrees-of-freedom (DOF) system development on one side and insufficient DOF for accurate simulation on the other. In this study, we evaluated the minimum DOF of thigh segment that a gait simulator should have to test the performance of prosthetic knees in a cost-effective manner. Initially, numerical simulations of swing phase of prosthetic leg with IITM polycentric knee (IPK) using 3D gait data and with different arrested DOF of the thigh were performed to identify the essential DOF of gait simulator. By comparing different cases of arrested DOF with the six-DOF ideal case, it was revealed that only sagittal plane movements, namely flexion-extension, vertical translation, and horizontal translation, are sufficient to test prosthetic knees. Subsequently, a compact and modular gait simulator was developed. Hardware-in-loop simulations of the IPK using the gait simulator were used to demonstrate its effectiveness in assessing the performance of prosthetic knees, which validated the ability of the IPK to extend completely without an extension assist before heel contact. Additionally, it was exposed that the IPK's extension stop needs redesigning to effectively absorb the impact energy when the knee extends completely before heel contact. These findings emphasize the significance of a cost-effective gait simulator in prosthesis development and reduce dependency on user trials.

1. Introduction

Lower limb amputations profoundly impact an individual's mobility and quality of life. Prosthetic knees play a critical role in restoring walking ability and stability during gait [1,2]. Consequently, evaluating the performance of prosthetic knees is essential for improving their functionality and ensuring user's satisfaction. Typically, user trials are carried out for feedback during the development stage of prosthetic knees. However, ethical approval and participant consent required for the trials make it a lengthy process and there could be unknown risks to the participants during trials. When combined with the necessity for numerous prototypes, this approach extends the product development time and increases costs.

Virtual engineering offers a promising approach to agile product development, enabling testing in virtual environments [3–6]. In prosthetic knee development, it reduces reliance on user trials, mitigating associated risks and liabilities, while also shortening development time [7,8]. It involves two stages: numerical simulations and hardware-in-loop simulations. In numerical simulation, computer-aided design (CAD) models of the prosthesis are created, and walking is simulated to predict the user's gait. Studies by Kim et al. [9], Colombo et al. [10,11], Xie et al. [12], and Lui et al. [13] have utilized numerical simulation to test new prostheses, enabling design refinement before physical prototyping. Similarly, we employed numerical simulations of swing phase of walking in our previous studies to develop the IITM polycentric knee (IPK) [14,15].

While numerical simulations provide valuable insights for design optimization, they are subject to assumptions made during modeling and may not fully account for real-world uncertainties, such as material properties, manufacturing constraints, and actuator settings. Thus,

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physical prototypes must be tested to ensure their performance aligns with simulation results and to address user's safety concerns. Furthermore, walking in complex, uneven environments—such as grassy or soil-covered terrain, commonly encountered by prosthesis users in developing countries—is difficult to simulate accurately using numerical methods. As a result, hardware-in-loop simulation, which involves testing physical prototypes under different simulated walking conditions, has become essential before user trials.

In hardware-in-loop simulations, robotic devices called gait simulators are used to replicate the walking patterns of prosthesis users [16–22]. An ideal gait simulator should provide six independent motions of the thigh in sagittal, frontal, and transverse planes, corresponding to the six degrees of freedom (DOF) of the residual thigh. Current gait simulators are designed to meet specific testing needs, such as testing control strategies, basic functioning and fatigue performance, and therefore, come with varying DOF [16–23]. For instance, Marinelli's simulator [16] with six DOF can test a prosthesis under complex three-dimensional movements, while others focus on specific planar motions [19,23] or are limited to treadmill walking [21,24]. Gait simulators with limited DOF and lacking certain motions may have difficulty in replicating the swing phase dynamics and analyzing different prosthetic knees [18,23]. Specifically, Borjain's gait simulator [23] could not simulate swing due to foot interference with the ground, attributed to the absence of hip vertical movement. Similarly, despite having 5 DOF, Thiele et al.'s gait simulator [18] experienced deviations in simulated knee angle during swing due to lack of horizontal hip movement. This drawback may also apply to other gait simulators with treadmill walking, which cannot replicate the shank acceleration-deceleration during normal walking scenario. Conversely, use of sophisticated and high DOF systems, such as biped robots [17,20] or industrial robots [16,25], could offer more biofidelic simulations, but often require significant capital investment, impacting the cost-effectiveness of prosthetic knee development. These gaps in the existing gait simulators underscore the need to analyze the essential DOF required for developing functional yet cost-effective gait simulators for hardware-in-loop simulations. Additionally, current gait simulator designs are tailored to specific testing needs and lack the flexibility to be upgraded for testing prostheses under complex walking conditions, limiting their ability to meet future testing requirements and support prosthesis development.

Therefore, the overarching objective of this study is to develop an affordable, efficient and modular (allowing for easy upgrades) gait simulator through a systematic development approach, thereby reducing dependency on user trials during prosthetic knee development. The current scope of the study is to address the existing testing requirements in our laboratory for affordability, while keeping the development process flexible to accommodate future needs. Specifically, we concentrated on the swing phase of polycentric knees, as the advantages of polycentric knees during stance phase are well established in the literature [2,26,27]. Our previous numerical simulations [14,15] indicated that the IPK exhibits complete extension during swing without requiring an extension assist, unlike a single-axis knee. Thus, our goal was to use the gait simulator to verify these findings under conditions similar to those utilized in the numerical simulations, before conducting user trials.

Before proceeding, it was necessary to determine the critical DOF in the hip joint that are essential for realistic swing phase simulations. Hence, this study encompassed four stages: 1) Identifying the minimum DOF required in the gait simulator, 2) Developing a gait simulator based on the identified DOF, 3) Evaluating the gait simulator's accuracy in physically simulating desired motion profiles, and 4) Validating the numerical simulation findings from our previous studies using the newly developed gait simulator.

2. Analysis for minimum degrees of freedom in the gait simulator

In our previous studies [14,15], 2D gait data from Winter [28] was used for analyzing prosthetic knees, potentially limiting the representation of movements in other planes during swing phase. To address this limitation for minimum DOF analysis, 3D simulations of walking with an IPK prosthetic leg were performed in this study. Gait trials of a healthy subject were conducted to capture three-dimensional gait data, which was then used as inputs for swing phase simulations using ADAMS software (Adams/View, 2017, MSC, Newport Beach, CA, USA). Different cases of swing phase of the IPK prosthetic leg, with each of the hip DOF arrested were performed and analyzed to identify the minimum significant motions that influence knee performance during swing.

2.1. Three-dimensional gait data for simulations

For the 3D simulations of walking with the IPK prosthetic leg, gait data of a healthy subject (age: 27 years, height: 170 cm, weight: 62 kg) collected at the Gait and Motion Analysis lab (GAMA lab), IIT Madras, was used. The data collection was part of a broader research initiative aimed at obtaining normative reference gait data to support the development of lower limb assistive devices. The study protocol included adults aged 18 years and older, with heights between 4 and 6 feet and weights between 30 and 100 kg. Pregnant women, children, elderly individuals, and those with neuromuscular dysfunctions that could affect normal walking patterns were excluded. The institutional ethics committee of IIT Madras approved the protocol (IEC/2017/04/SS-1/04). For the DOF analysis, we used 3D data of a subject who walked at a self-selected walking speed of 0.9 m/s, which falls within the range typically observed in transfemoral amputees (0.6 m/s to 1.1 m/s [29,30]).

Gait data was collected using PhaseSpace™ (PhaseSpace Inc., USA) motion capture system. The motion capture system comprised eight infrared cameras placed around a 15 m diagonal walking arena. The motion capture system acquired the gait data at 240 frames/s. A total of 35 active markers were used for acquiring three-dimensional gait data. The marker protocol used in this work was adapted from Robertson et al. [31] and is presented in Fig. 1. After practice walk with the markers, a total of five walking trials were recorded to ensure that at least one data with all the markers visible is captured. The human and prosthetic leg models used for 3D gait data analysis and simulations are presented in Fig. 2.

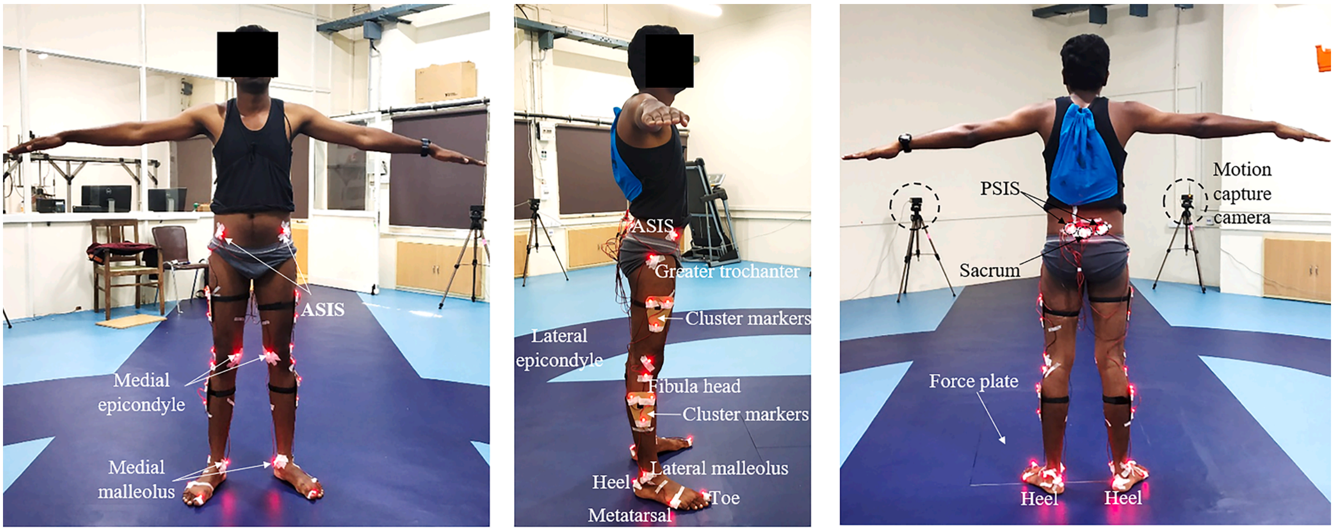
High-frequency noise in the raw marker data from the motion capture system was filtered using a low-pass, two-way, 2nd order Butterworth digital filter with cutoff frequency 6 Hz. The filtered marker data along with the force plate data were then applied to the lower body model of the healthy subject (Fig. 2b) to compute translational and rotational motions of the thigh during swing phase of walking. These motions are presented in Fig. 3. All motions are from the same walking trial to recreate the exact combination of thigh movements of the subject during simulations. These thigh motions were used as input motions for the prosthetic leg model in ADAMS.

2.2. Gait parameters studied

The success of swing phase with a prosthetic knee relies on factors such as knee angle and toe and heel clearances [14]. Hence, a comparison of knee angles, toe and heel positions for different cases of DOF at the hip was performed to identify the minimum DOF for a gait simulator that can test prosthetic knees.

2.3. Prosthetic leg model and constraints for simulations

Matching the dimensions of the leg of the healthy subject (greater trochanter height: 0.877 m, lateral epicondyle height: 0.473 m, lateral



a) Front view **b) Side view** **c) Rear view**

Fig. 1. Marker protocol used for 3D gait analysis.

malleolus height: 0.060 m), a prosthetic leg with IPK was modeled in ADAMS (Fig. 2d). The method reported in Sudeesh et al. [14] was followed to estimate the body segment parameters (BSP) of the prosthetic leg, such as mass, moment of inertia, and center of mass. The estimated BSP are provided in Table 1. The IPK is a passive knee with only frictional damping. A brass bush compressing against a steel pin at joint K1 (Fig. 2d) controls the damping level in IPK. In Adams, preload torque applied at joint K1 emulates frictional damping as applied in the physical IPK [14]. The preload torque that gives maximum knee extension before heel contact was considered as the appropriate preload torque, as that is a criterion for stable knee at heel contact.

For a given frictional damping at the knee, the knee motion depends on the state of motion of the leg at toe-off (initial position and velocity) and the thigh movements during swing [32,33]. Normative data for hip position and orientation of the thigh at toe-off (0 % swing phase in Fig. 3) were applied as the initial conditions for the swing phase simulations. However, for a normal hip position and thigh orientation at toe-off applied to the IPK leg, the foot did not touch the ground as polycentric knees tend to shorten the leg during flexion. Therefore, the shank orientation had to be adjusted in such a way that the foot is just leaving the ground, as was done in Sudeesh et al. [14,15]. The initial velocity (velocity at toe-off) of the prosthetic leg was computed assuming that the foot leaves the ground like the healthy leg. In other words, normal hip motions and knee motions during a complete gait cycle (stance + swing) were applied to the prosthetic leg for a kinematic analysis to compute its velocities at toe-off for simulations. It should be noted that a gait simulator with reduced DOF does not allow all the six motions of the thigh. Therefore, only those motions over which the simulator has control were used while computing the initial conditions in this study.

Once initial conditions were applied, three-dimensional thigh motions (Fig. 3) were used to simulate swing phase of walking with the prosthetic leg. As mentioned before, an ideal gait simulator should have six DOF at its mechanical hip joint. Hence all six thigh motions were first applied to the prosthetic leg model for swing phase simulations, and gait parameters obtained from it were taken as reference. Then, each DOF at the hip was arrested. The corresponding simulated motion of the lower limb was compared against the reference for analyzing the influence of thigh motions on the performance of the knee. Thigh flexion-extension is a compulsory movement for walking, and thus it was not arrested. The DOF required at the hip joint in a gait simulator to test various prosthetic knees were identified based on the deviations in the computed knee angles and foot positions during swing.

2.4. Numerical simulation results and DOF analysis

The appropriate preload torque for frictional damping that provided maximum knee extension for the IPK leg was found to be 1 Nm. With 1 Nm preload torque, the swing phase was simulated for the prosthetic leg with all six DOF, and for cases with different arrested DOF at the hip. The comparison of knee angles from simulations (Fig. 4a) indicated that the motions other than hip flexion-extension do not have much influence on the knee motion during swing. In comparison to other thigh motions, the sagittal plane translations (horizontal (x) and vertical (y) movement) were found to have greater influence on the knee angle (Fig. 4b).

Arresting hip translations (x, y, and z) affected the position of the foot in the corresponding direction (Fig. 4c-h). It was also observed that the abduction/adduction and internal-external rotation notably affect only the medial-lateral (z) positioning of the foot (Fig. 4e, h). Since the medial-lateral position of the foot does not affect the success of swing (foot interference with the ground), they are not considered as essential DOF in this study.

It was observed that the toe interfered with the ground in the absence of vertical movement (y) of the hip. For all the cases studied here, heel interference with the ground occurred during terminal swing, a unique characteristic of the IPK resulting from the posterior translation applied for increased stability during stance (as described in Sudeesh et al. [14, 15]). Prosthesis users might opt for a hip-lifting strategy, which involves the vertical movement of the hip, to achieve foot clearance during swing [14]. Henceforth, vertical movement is considered an essential DOF for a gait simulator.

In the absence of horizontal movement, the maximum deviation in the knee angle was found to be 3°. The gait simulator developed by Thiele et al. [18] also observed divergence in the knee angle as there was no provision for horizontal movement during swing. Thus, the horizontal movement might significantly influence the knee motion for a different walking speed or pattern. Additionally, when a user trips during swing phase, the upper body falls forward, placing a load on the knees. Some prosthetic knees come with safety features such as C-Leg's stumble recovery [34] or IPK's 2° stable knee flexion [27]. To test such safety features, the horizontal movement that depicts the trunk motion is required. More importantly, to achieve realistic toe-off condition for swing phase simulations, the gait simulator needs to simulate the late stance phase as well. For stance phase, there should be a relative

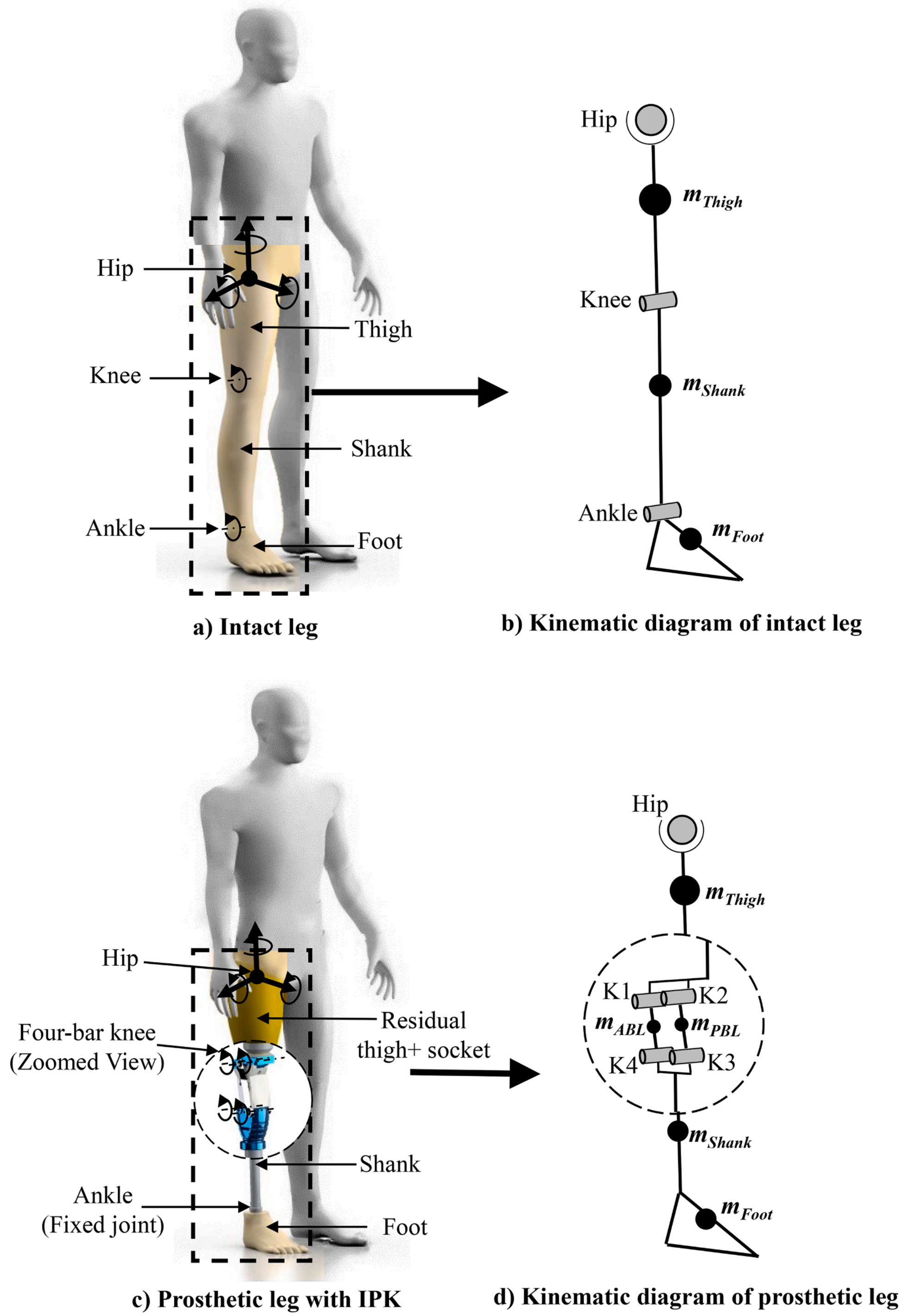


Fig. 2. Intact and prosthetic leg models used for 3D analyses.

horizontal movement between hip and ground. Hence, we concluded that a gait simulator to test prosthetic knees close to realistic walking scenarios should have at least three DOF: flexion-extension, horizontal and vertical translations.

3. Development and testing of gait simulator

Based on the numerical simulation results, we then developed a gait simulator capable of providing thigh flexion-extension, horizontal and vertical translational movements. A systematic four-phase approach was used for the development of the gait simulator: 1) conceptual design phase, 2) detailed design phase, 3) prototyping and 4) testing.

3.1. Conceptual design of gait simulator

A compact and modular design was conceived for the gait simulator. Compactness saves space and simplifies the operation, while the modular concept provides provision for future modifications, such as adding additional DOF. It is important to note that the gait simulator should also simulate late stance to achieve a realistic toe-off condition for swing phase simulations. Therefore, simulation of late stance phase was also considered during the development of the gait simulator.

The kinematic chain of the proposed gait simulator is presented in Fig. 5. Simultaneous motion of three independently moving parts (thigh segment, table, and saddle) of the gait simulator can simulate the

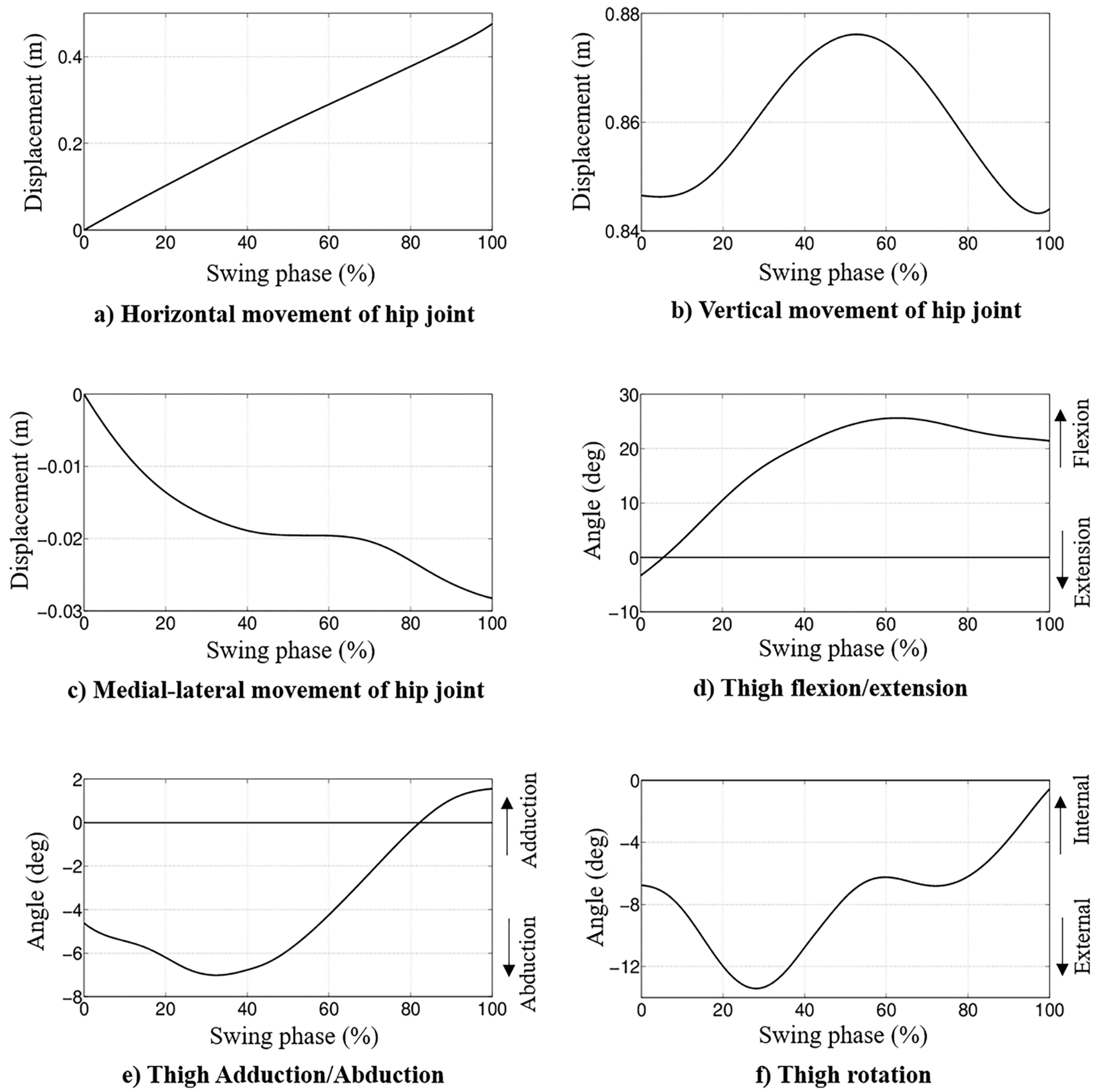


Fig. 3. Thigh motions of the able-bodied person used for the swing phase simulation of the prosthetic leg.
Note: 100 % swing phase corresponds to 0.48 s in time scale.

Table 1
Body segment parameters of the prosthetic leg used in the simulation.

Leg segment	length (m)	Mass (kg)	Centre of mass* (m)			Moment of inertia ($kg.m^2$)		
			x**	y	z	x	y	z
Thigh	0.403	8.78	–	0.165	–	0.1615	$3.31(10^{-2})$	0.1615
ABL	0.082	0.096	–	0.041	–	$1.27(10^{-4})$	$3.62(10^{-5})$	$9.44(10^{-5})$
PBL	0.076	0.060	–	0.038	–	$3.8(10^{-5})$	$4.77(10^{-6})$	$4.02(10^{-5})$
Shank	0.333	0.446	–	0.151	–	$7.15(10^{-3})$	$1.14(10^{-4})$	$7.16(10^{-3})$
Foot	0.231	0.664	0.053	0.041	–	$7(10^{-4})$	$2.7(10^{-3})$	$3.4(10^{-3})$

Abbreviations: ABL, Anterior Binary Link; PBL, Posterior Binary Link.
* From the proximal joint for various link segments.
** x is along walking direction; y is along vertical direction and z is along medial-lateral direction.

sagittal movements of the transfemoral prosthesis user's thigh. The thigh segment is connected to the table through a pin joint (acting as hip joint), which allows flexion-extension. The table is connected to the saddle through a prismatic joint (linear bearings), allowing only vertical movement between them. The movement of the table simulates the vertical translation of the transfemoral prosthesis user's hip. For uniformity and compactness, all motions were chosen to be controlled using servo motors. A ball screw was used to convert the servo motor rotational motion to the vertical movement of the table. The horizontal translational motion was achieved through another prismatic joint, which lies between the saddle and frame. The walking distance or step length depends on the range of motion of the saddle. It is necessary for the saddle to have a movement range of approximately 1.5 m to effectively test the prosthesis throughout a complete gait cycle [28]. A timing belt and pulley mechanism was found to be a suitable option to move the saddle over such long distances. When rotational motion is applied to

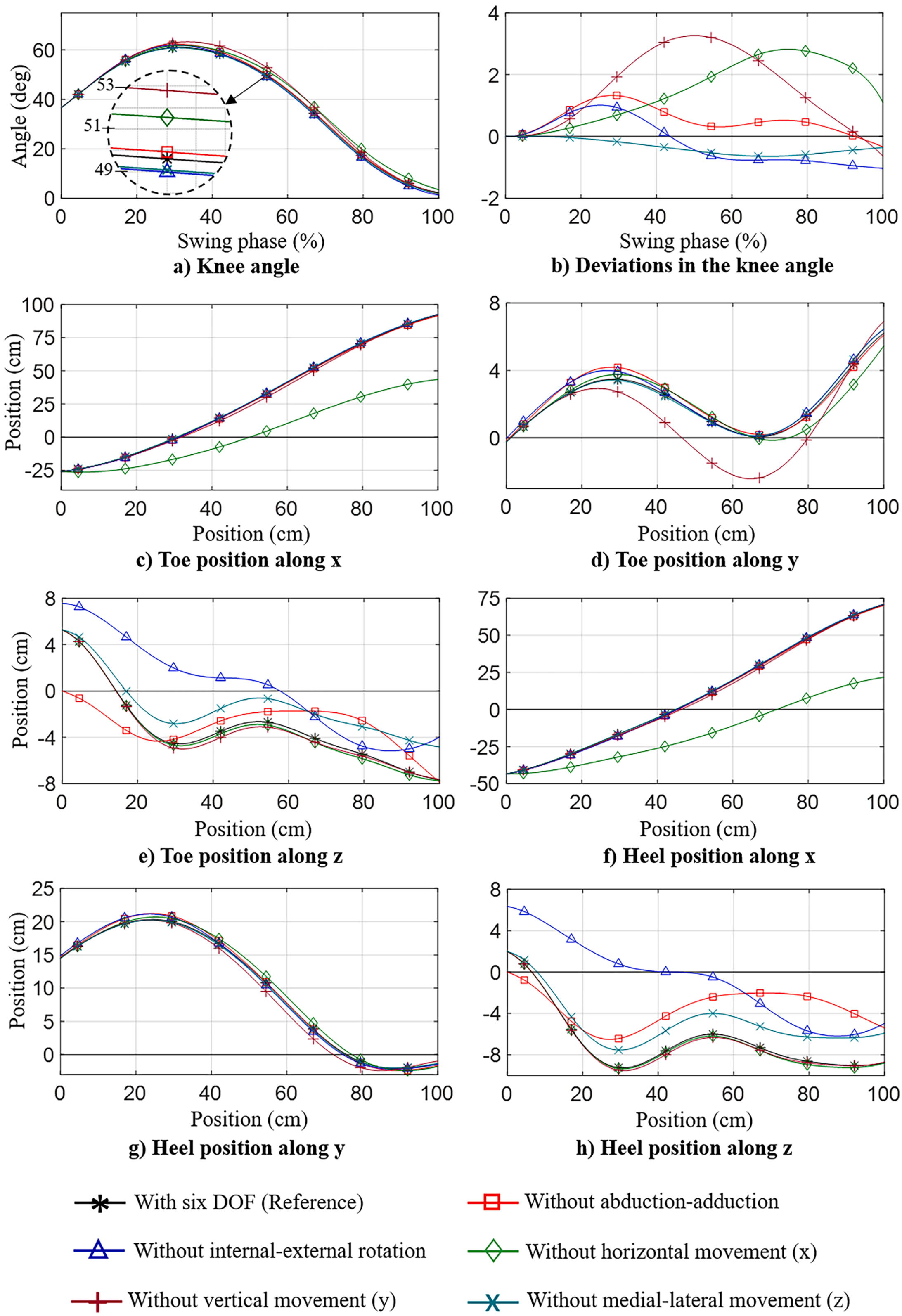


Fig. 4. Comparison of gait parameters for the cases with different arrested DOF of the gait simulator opt for a hip-lifting strategy, which involves the vertical movement of the hip, to achieve foot clearance during swing [14]. Henceforth, vertical movement is considered an essential DOF for a gait simulator.

one of the pulleys, the timing belt attached to the saddle causes horizontal movement. The base acts as the ground to test the prosthesis during stance as well.

3.2. Detailed design and specification of the gait simulator

Following the conceptualization, we entered the detailed design phase of the simulator, in which individual components were selected based on performance requirements and design specifications. The

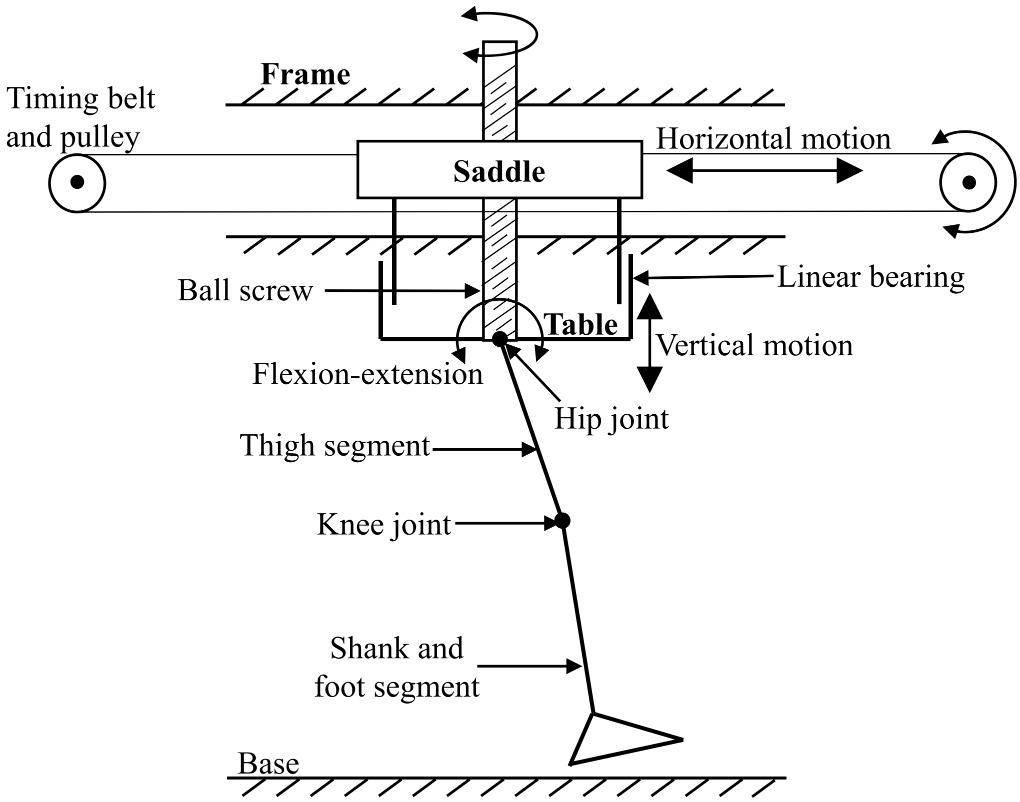


Fig. 5. Kinematic diagram of gait simulator with three degrees of freedom.

detailed design of the gait simulator is presented in Fig. 6. The frame of the gait simulator was made of aluminum extrusion channels to reduce weight and realize modularity for adding more DOF. A gait simulator should be able to test a prosthesis under fast-walking scenarios if required. A walking speed (v) normalized by the height of the person (h) above 1 (i.e., $v/h > 1$) is considered as fast walking [35]. Hence, hip kinematics and kinetics during fast walking were taken as the basis in the design. The maximum hip angular velocity required even for fast walking is only 60 rpm [35]. On the contrary, the maximum hip moment is nearly 80 Nm during fast walking. Hence, a servo motor (1800 rpm, 4 Nm rated torque) with gear reduction (1:20) was required to achieve high torque at low rpm. The vertical motion was controlled through another motor coupled to a ball screw (20 mm lead), which converts the rotational motion into linear motion of the table. After considering the vertical velocity during fast walking and the mass of the table along with the prosthesis (36 kg), the motor requirement—4.2 Nm torque at 1140 rpm—for the vertical motion was determined. The saddle has V-grooved wheels that engage with V-shaped rails (aluminum extrusion channels) on the frame, forming a prismatic joint, as shown in Fig. 6. A timing belt attached to the saddle provides horizontal motion with the rotation of the pulley (diameter: 76.9 mm). The pulley rotation was controlled using a motor. The motor for horizontal motion must move a total mass of 110 kg, including that of the saddle, table, and prosthesis. Therefore, it was found that the pulley should receive a maximum input torque of 38 Nm and a maximum speed of 480 rpm. Thus, a motor with a rated torque of 7.71 Nm at 3200 rpm in conjunction with a gear reducer (1:5) would suffice for horizontal motion.

The residual thigh has a complex geometry and depends on the individual anthropometry and amputation characteristics. Additionally, the socket design affects its fit. Therefore, like in our previous numerical simulations[14,15], we have treated the residual thigh together with the socket as a single rigid link, blocking the influence of variations in both residual thigh and socket. Two pipes, one with 30 mm outside diameter and another with 33 mm inner diameter and a slit connected with a hose clamp formed the thigh segment to which the prosthetic knee was attached to the gait simulator (Fig. 6). The smaller pipe can slide inside the larger pipe to vary the thigh segment length. This way, the gait simulator also provides the flexibility to test the prosthetic knee for people of different heights, thereby expanding its utility to a broader spectrum of users.

3.3. *Prototyping*

A prototype gait simulator (Fig. 7) was then fabricated based on the design outlined in the previous section. The development of the gait simulator involved integrating and synchronizing different subsystems, including mechanisms for each DOF, safety features, servo drives, and software interface. Due to the capital investment required for compact and high-power servo drives, a step-by-step approach was adopted to ensure cost-effectiveness. Initially, mechanical subsystems were fabricated and tested for efficacy. The integration of motors for each DOF was carried out progressively, starting with flexion-extension, followed by vertical and horizontal movements. Labview software was utilized to control the servo drives, which received analog input signals to simulate various walking scenarios. Considering the substantial capital investment involved in acquiring servo drives, the prototype utilized motors with reduced power to primarily evaluate their effectiveness in testing prosthetic knees before advancing to full-scale development. This incremental approach allowed for a more rational investment strategy, providing ample time for the development of the prosthetic knees and the associated testing equipment. Flexion-extension was controlled using a brushless motor with a rated torque of 0.65 Nm and 3000 rpm, with a gear reducer (1:50) to provide the thigh segment with 32.5 Nm at 60 rpm. For horizontal motion, a brushless motor with a gear reduction provided 2.1 Nm torque and a maximum speed of 600 rpm. However, compromise was not made for the vertical motion, as it plays a critical role in controlling the table's movement under gravity and ground reaction forces (GRF). Therefore, an AC servo motor capable of delivering

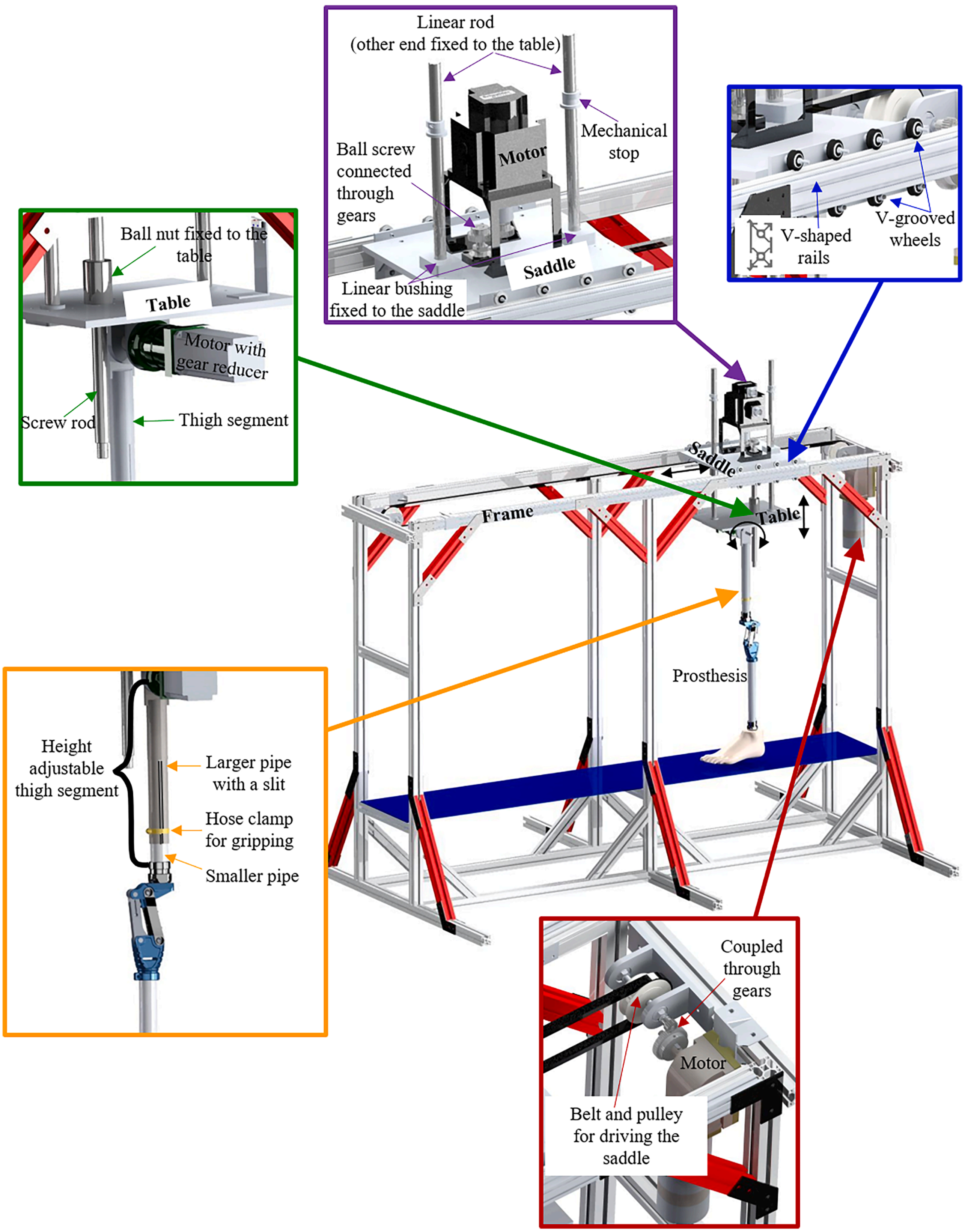


Fig. 6. Modular design of the gait simulator.

4.2 Nm torque at 1800 rpm was used for vertical movement.

3.4. Testing of the gait simulator

The servo drives were integrated into the gait simulator, and their reliability in delivering desired motion profiles to the prosthesis was

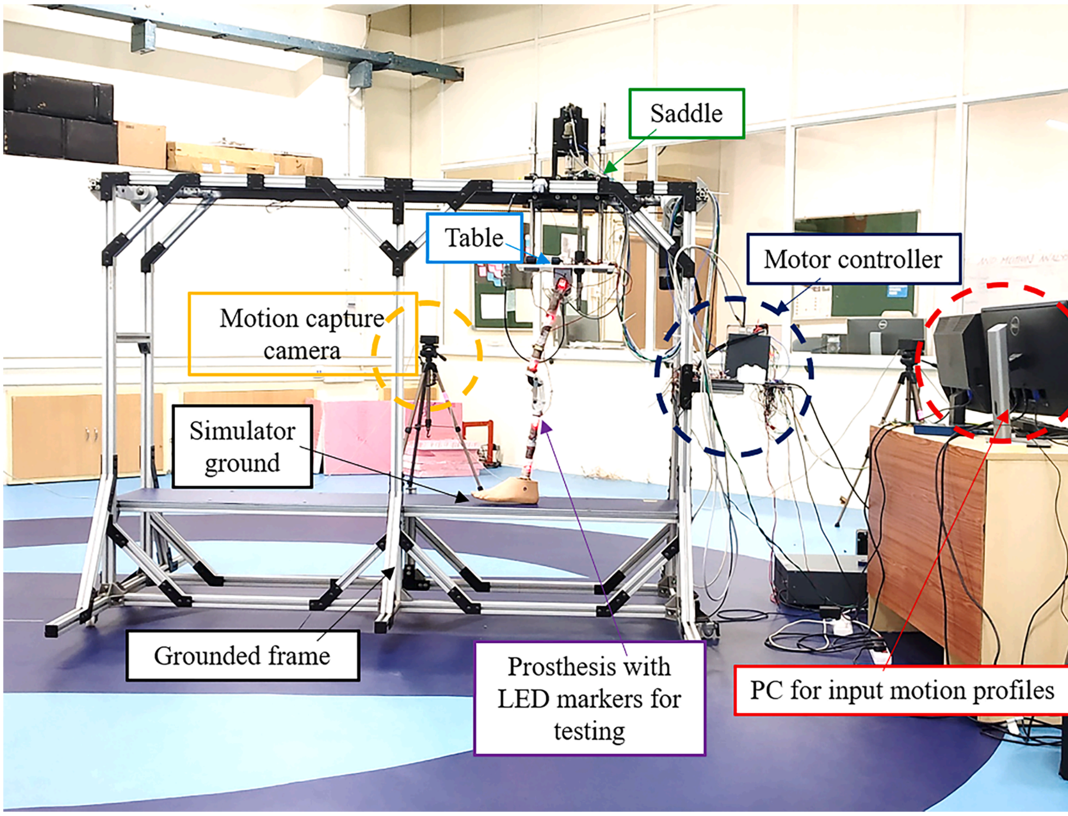


Fig. 7. Prototype of the gait simulator.

tested. To achieve a realistic toe-off condition (velocity of the prosthetic leg at toe-off) during experimentation, the gait simulator needed to simulate a part of the stance phase. Thus, the servo drives underwent testing over a complete gait cycle encompassing both stance and swing phases. Since the replication of the desired thigh motion was the focus, prosthetic leg was not connected at this stage, and only the motion of the thigh segment (outer pipe with slit in Fig. 6) was considered. One of our objectives was to verify the simulation findings in our previous works [14,15]. To accomplish this, motion profiles obtained from Winter [28] were applied to the gait simulator, and the resulting motion profiles of the thigh segment (flexion-extension, horizontal, and vertical translations) were captured using the same motion capture system described in Section 2.1. These captured profiles were then compared against the input motion profiles for analysis.

3.4.1. Comparison of input and output motion profiles of the gait simulator

Thigh motions generated by the gait simulator corresponding to the input motion profiles are presented in Fig. 8. The gait simulator reproduced the vertical movement quite closely (Root Mean Square Error (RMSE): 0.8 mm). However, significant progressive errors were observed in horizontal movement (RMSE: 300 mm; Max error: 430 mm) due to limitations in the motor's ability to handle rapid acceleration and deceleration of the heavy saddle. Although variations were noted in flexion-extension (RMSE: 2.3°), the overall shape of the output thigh angle closely followed the input.

As mentioned earlier, the horizontal movement is significant while testing the prosthetic knee safety features or stance phase performance. Furthermore, it was observed that the major deviations in flexion-extension from the reference value (Fig. 8c) were during stance phase, which comprises 0 - 60 % of the gait cycle. The maximum deviation in flexion-extension during swing was only 2° Thus, the current form of the gait simulator was considered suitable to verify the ability of the IPK to extend even without an extension assist during swing, as presented in the following section.

4. Testing of IPK using gait simulator

Following the development of the gait simulator, we used it to test a physical prototype of the IPK by applying normal thigh motions [28] used in our previous numerical simulation studies [14,15]. Stance phase was included solely to achieve realistic toe-off conditions during swing phase analysis. The experimental setup shown in Fig. 7 was used. It should be noted that the motor used for flexion-extension was under-powered to simulate stance phase, which requires a higher moment. Therefore, by slightly lifting the hip and reducing the pressure on the floor during stance, the load on the motor was minimized. This adjustment eliminated the need for a hip-lifting strategy required for foot clearance during swing with the IPK [14]. Thus, this experiment focused solely on evaluating the IPK's ability to extend without an extension assist. We repeated the experiment with the same input conditions and hip lifting adjustment to check our results. Thigh, shank, and knee angles were the gait parameters analyzed here. For comparison, we include the numerical simulation results of the IPK presented in Sudeesh et al. [14].

4.1. Hardware-in-loop simulation results

During hardware-in-loop simulations, it was observed that the thigh angles generated by the gait simulator closely matched the required angle (Fig. 9a). Both the numerical simulation and the gait simulator exhibited increased shank angles at toe-off compared to normal walking (Fig. 9b), attributed to the shortening effect of the polycentric knee [14]. The gait simulator showed higher shank angles at toe-off than the simulation by 11° (trial 1) and 13° (trial 2), resulting from the hip being raised to reduce floor pressure during stance phase simulation using a low-powered motor. This adjustment led to increased knee extension

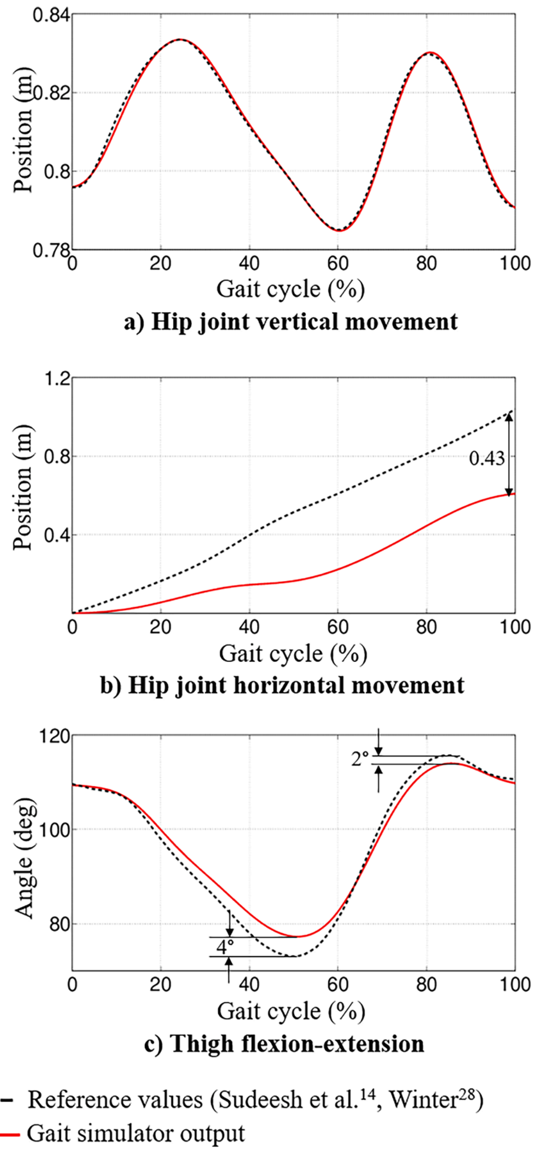


Fig. 8. Comparison of input and output motion profiles of the gait simulator
Note: 100 % gait cycle corresponds to 0 .99s.

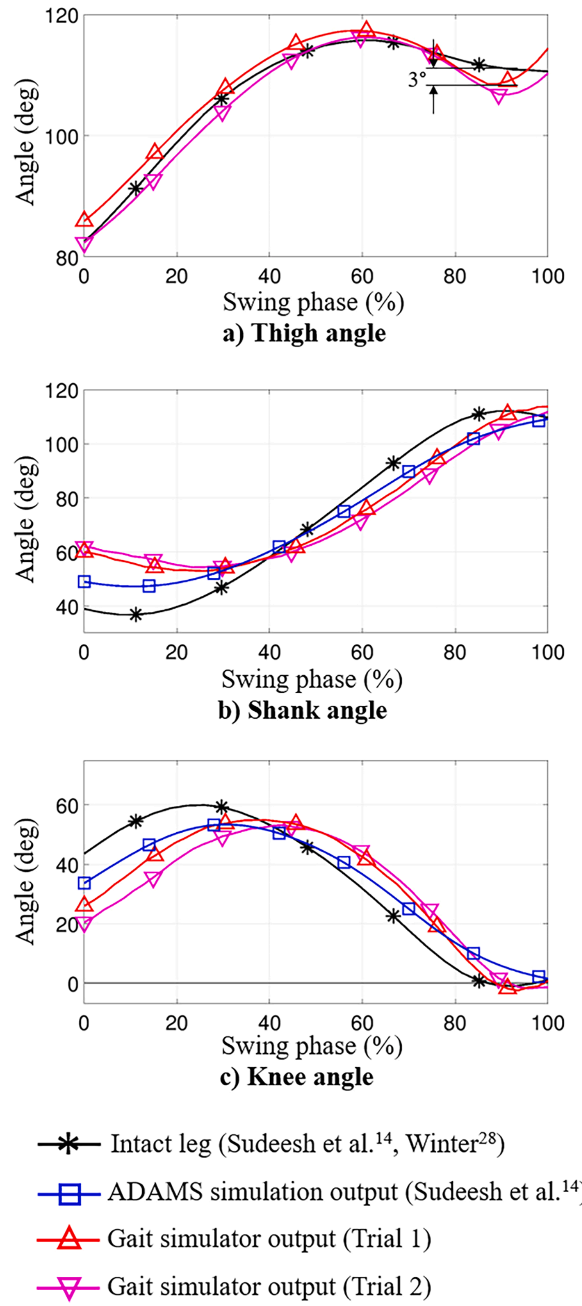


Fig. 9. Swing phase analysis of the IPK using gait simulator.

(Fig. 9c), allowing the foot to maintain contact with the ground during pre-swing. Nonetheless, the main objective of the gait simulator testing was to verify the IPK's ability to achieve complete extension before heel contact. The knee angle of the IPK obtained from the gait simulator supported the simulation findings, demonstrating that the IPK can fully extend even without an extension assist. Notably, a sudden thigh flexion was observed for both trials at the end of swing (Fig. 9a), attributed to the impact of the extension stop at full knee extension, indicating the need to redesign the extension stop to absorb the impact force effectively.

5. Discussion

Cost-effective and efficient methods for prosthesis testing are essential to the agile development of affordable prosthetic legs. The development approach for the gait simulator presented in this study emphasizes both cost-effectiveness and efficacy through a systematic, multistage process. An initial DOF analysis, conducted via numerical simulations, helped reduce unnecessary complexity and irrational investment for us. For instance, Marinelli's gait simulator, which uses a 6-DOF industrial robot paired with a 2-DOF moving platform acting as the ground, requires significant capital investment and considerable laboratory space—making it excessive for prosthetic testing purposes. In contrast, our 3-DOF gait simulator is more compact, less complex, and requires lower capital investment while still allowing biofidelic simulations of swing phase of walking.

Vertical movement in our simulator is crucial for analyzing foot clearance, while horizontal movement captures the effects of acceleration and deceleration in the walking direction. Additionally, the flexion-extension capability allows for evaluation of the hip effort needed to walk with a prosthesis, as demonstrated in our previous work [14,29]. Together, these three DOF allows our gait simulator to test prosthesis under typical normal walking conditions.

A limitation of this study is the lack of statistical power due to limited gait data diversity. Our validation is based on data from a single subject's walking pattern. Other DOF may have a significant impact for different walking profiles. To address this in the future development, we designed the gait simulator to be modular, allowing for additional DOF to be integrated if necessary for testing more complex walking scenarios, such as circumduction and quick forward or lateral falls.

The gait simulator demonstrated high accuracy in simulating the vertical motion of the hip, unlike the other two motions (Fig. 8). This level of accuracy (maximum error: 1.5 mm) is better than values reported in the literature (12 mm [7]). Furthermore, the accuracy of the flexion-extension motion (maximum error: 2° during the swing phase) with the lower powered actuator remains comparable to that reported by Xu et al. [7] and considerably better than other simulators (maximum error $> 7^\circ$ [18,19]). With appropriately rated actuators, as described in Sec 3.2, the gait simulator can be further developed to provide even more accurate simulations of walking under different scenarios.

Due to certain limitations, such as the limited actuator power rating and the current inability to simulate complex walking movements, the gait simulator is still in the preliminary stages of development. Nevertheless, the development methodology employed in this study met our current requirements cost-effectively by validating the prosthetic knees joint's (IPK) ability to extend completely without an extension assist. Additionally, the deficit in impact absorption characteristics of the extension stop were not captured during our previous numerical simulation studies [14,15], underscoring the importance of hardware-in-loop simulations to ensure user safety and avoid multiple design iterations.

A potential direction for future development is the incorporation of appropriate actuators for horizontal movement and flexion-extension, along with force-measuring transducers to compute the ground reaction forces. The base of the gait simulator can also be adapted to accommodate different walking surfaces, such as soil, uneven terrain or grass, simulating a variety of real-world environments. This would enable simulations of stance phase and other complex movements in diverse conditions. Furthermore, the current gait simulator cannot study the influence of socket design and characteristics of the residual thigh. Future studies will address this limitation by incorporating biofidelic residual thigh segments and sockets of various sizes and shapes.

6. Conclusions

The practical use of a gait simulator for testing physical prototypes of prosthetic knees has been limited by insufficient DOF on one side and the complexity and substantial capital investment on the other. The numerical simulations conducted in this study identified that three sagittal plane motions—hip vertical and horizontal translations and flexion-extension—are the essential DOF, while other motions have minimal impact on knee motion during swing phase of walking. Based on these findings, a compact, modular, and cost-effective gait simulator was developed for testing prosthetic knee prototypes using a systematic, multistage process.

The gait simulator proved valuable for prosthesis development through hardware-in-loop simulations of the IPK leg, successfully validating its remarkable ability to extend completely without the need for an extension assist mechanism during swing. This contributed to the development of a simplified, cost-effective, and functional prosthetic knee for transfemoral amputees.

Further development of the gait simulator, incorporating additional DOF, force-measuring transducers and biofidelic residual thigh segments to evaluate the performance of the knee, foot and socket under more complex movements, could revolutionize prosthesis development. This study lays the foundation for such future progress in the field.

Competing interests

The author(s) declare no potential conflicts of interest with respect to the authorship, and/or publication of this article.

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Ethical approval

The institutional ethics committee of IIT Madras approved the human subject data collection protocol used in this study (IRB No: IEC/2017/04/SS-1/04).

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